

**Raw enwik9 dataset
1,000,000,000 bytes**

Main model evaluation

Independent 2,048-byte chunks
ASCII mapping

Llama 2 7B
full-vocabulary logits

Accumulated next-token
negative log-probabilities

BPC calculation and
high-bit correction

**8.8967% paper-comparable
model-based estimate**

Practical coding validation

First 1,024,000 bytes
500 independent 2,048-byte chunks
ASCII mapping

Llama 2 7B
sequential token prediction

Top-100 categories plus escape
and exact token-ID stream

Range-coded archive
with metadata and high-bit mask

**10.5355% actual archive
byte-exact reconstruction verified**

Conventional baselines

Complete raw enwik9
no chunking or ASCII mapping

gzip level 9, bzip2 level 9,
LZMA2/XZ preset 9

Actual compressed
archive files

Independent decompression
and checksum comparison

**gzip 32.2592%
bzip2 25.3978%
LZMA2/XZ 21.3371%**